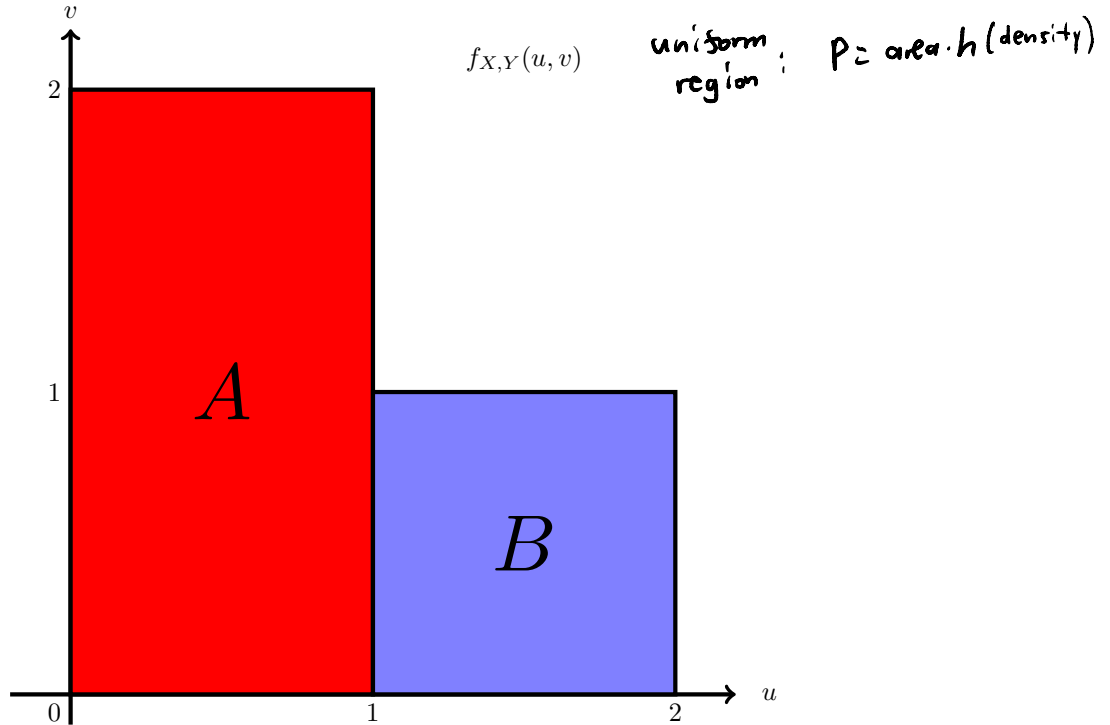


Suppose random variables X and Y have a joint probability density function $f_{X,Y}(u, v)$ which equals A in the red region, B in the blue region, and 0 elsewhere, as shown in the figure. Suppose the joint cumulative distribution function is $F_{X,Y}(u, v)$. If $F_{X,Y}(1, 3) = 1/2$, then what is $F_{X,Y}(3/2, 3/2) - F_{X,Y}(1, 1)$?



$$1 = 2A + B$$

$$F_{X,Y}(1, 3) = P(X \leq 1, Y \leq 3) = 2A = \frac{1}{2}$$

$$\rightarrow A = \frac{1}{4}$$

$$\rightarrow 1 = \frac{1}{2} + B$$

$$\rightarrow B = \frac{1}{2}$$

$$F_{X,Y}(1, 1) = A = \frac{1}{4}$$

$$F_{X,Y}\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{3}{2}A + \frac{1}{2}B = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

$$\rightarrow F_{X,Y}\left(\frac{3}{2}, \frac{3}{2}\right) - F_{X,Y}(1, 1) = \frac{3}{8}$$

(a) $3/8$

(b) $3/4$

(c) $3/16$

(d) $1/8$

(e) $1/4$

(f) $1/2$

(g) $5/16$

(h) $7/16$

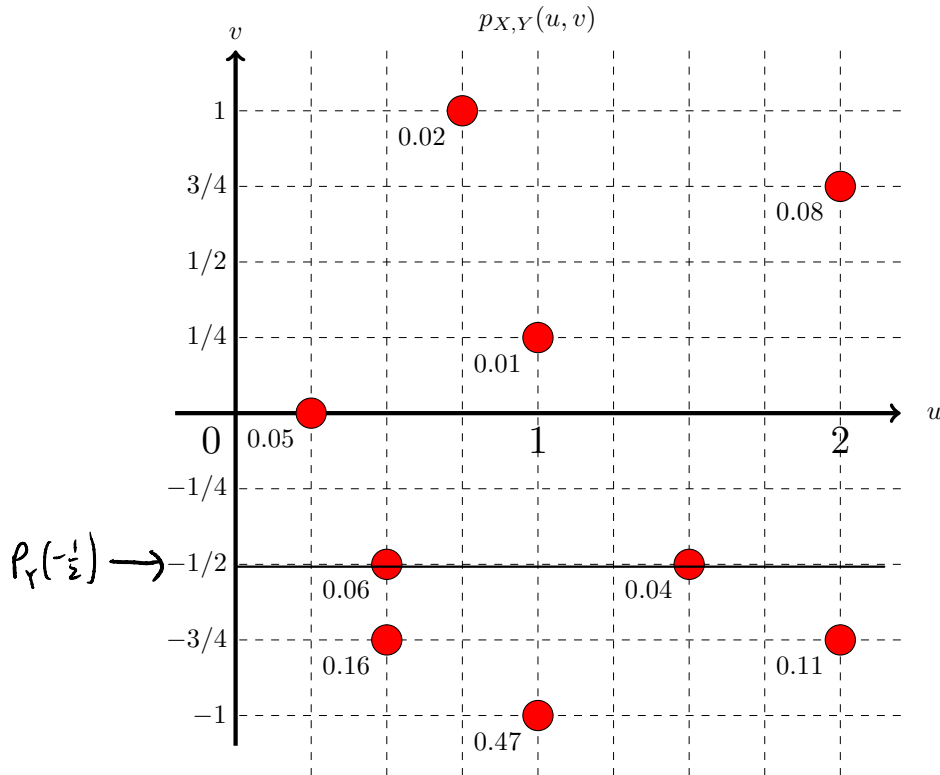
(i) $1/32$

(j) $3/32$

(k) $5/32$

(l) None of these.

Random variables X and Y have the joint probability mass function $p_{X,Y}(u, v)$ shown below, joint cumulative distribution function $F_{X,Y}(u, v)$, and marginal probability mass functions $p_X(u)$ and $p_Y(v)$. What is the value of $p_Y(-2^{-1}) + F_{X,Y}(2^{100}, 2^{-100})$?



$$p_Y(-2^{-1}) + F_{X,Y}(2^{100}, 2^{-100})$$

(a) 0.99

(b) 0.90

(c) 0.89

(d) 0.94

(e) 1.11

(f) 0.98

(g) 0.1

(h) 1.08

(i) 0.12

(j) 1.37

(k) 0.06

(l) None of these

$$p_Y(-2^{-1}) = p_Y(-\frac{1}{2}) = .06 + .04 = 0.10$$

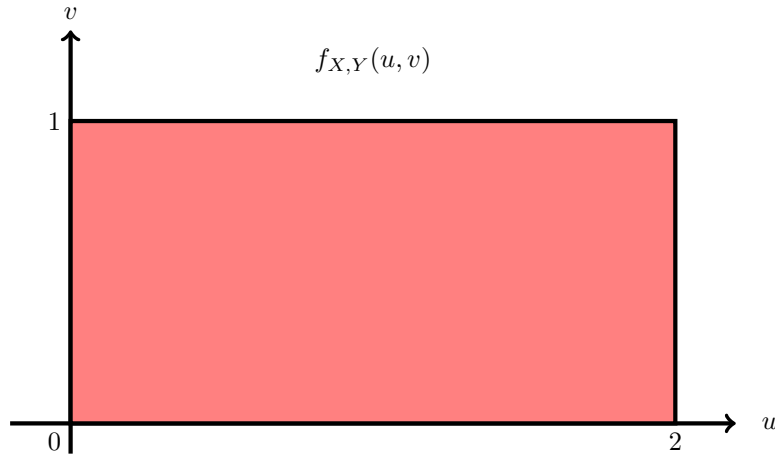
$$F_{X,Y}(\sim \infty, \sim 0) = P(X \leq \infty, Y \leq 0)$$

$$= .05 + .06 + .04 + .16 + .11 + .47$$

$$= .89$$

$$.10 + .89 = .99$$

Suppose random variables X and Y have a joint probability density function $f_{X,Y}(u, v) = uv$ in the red region, and zero elsewhere. What value of A would make the probability that Y is less than A equal to $1/2$?



(a) $\sqrt{2}/2$

(b) $1/16$

(c) $1/2$

(d) $1/4$

(e) $1/8$

(f) $3/8$

(g) $5/8$

(h) $1/\sqrt{8}$

(i) $\sqrt{2}$

(j) $1/3$

(k) $2/3$

(l) None of these.

$$f_{X,Y}(u,v) = uv$$

$$P(Y < A) = \frac{1}{2}$$

$$\rightarrow \frac{1}{2} = \int_0^A \int_0^2 uv \, du \, dv$$

$$= \frac{1}{2} \int_0^A v \left[u^2 \right]_0^2 \, dv$$

$$= \frac{1}{2} \int_0^A v (4) \, dv$$

$$= \cancel{2} \cdot \frac{1}{2} \left[v^2 \right]_0^A$$

$$\frac{1}{2} = A^2$$

$$\sqrt{\frac{1}{2}} = A$$

$$\frac{\sqrt{2}}{2} = A$$